

Indian Institute of Engineering Science and Technology, Shibpur
Department of Computer Science and Technology

Algorithm Lab (CS2271)
Assignment 4

1. An online **educational analytics platform** conducts **national-level mock examinations** for various competitive tests. In each examination session, approximately **110,000 students** participate simultaneously. Once the test concludes, the scores obtained by all students are collected and stored on the server in the form of an **unsorted integer array**. To generate real-time performance analytics, such as identifying the central tendency of student performance, the platform needs to compute the median score as efficiently as possible. Since the dataset is large, the choice of algorithm has a significant impact on both execution time and memory utilization. Write a C program to determine the median score using two different approaches, as described below:

- i) **Fixed Pivot-Based Approach:** Employs a divide-and-conquer approach in which the pivot is selected in a fixed manner.
- ii) **Quickselect-Based Approach:** Determine the median using the Quickselect algorithm, which finds the required order statistic without fully sorting the array.

After implementing both methods, perform a comparative analysis of the two approaches by evaluating and discussing their:

- i) **Time complexity** in the best case, average case, and worst case
- ii) **Space complexity** in the best case, average case, and worst case

Finally, justify which approach is more suitable for handling large-scale, real-time examination data and explain your conclusion with appropriate reasoning.

2. A **national digital census and survey platform** periodically gathers extensive **demographic and economic information** from **120000 households** distributed across the country. Among the various data points collected, **household income values** are stored on the server in the form of a large **unsorted array**. After the completion of each survey cycle, policymakers and analysts require reliable statistical indicators to support **data-driven decision-making**. One of the most critical measures is the **median household income**, as it provides a robust representation of central tendency and is not unduly influenced by extreme income values.

Given the **enormous size of the dataset**, sorting the entire array to compute the median becomes **computationally expensive** and **impractical** for real-time or near real-time analysis. To overcome this limitation, the analytics team opts for a **deterministic linear-time selection algorithm**, commonly referred to as the **Median of Medians method**, which guarantees **$O(n)$** worst-case time complexity.

Write a **C program** that computes the **median household income** using the **Median of Medians algorithm**. The implementation should avoid full-array sorting and ensure deterministic performance irrespective of input distribution.

After successfully implementing the algorithm, perform a **comparative performance analysis** between the **Median of Medians method** and the **Quickselect-based median finding method**. The comparison must include a detailed evaluation of both algorithms in terms of their:

- i) **Best-case, average-case, and worst-case time complexities**
- ii) **Best-case, average-case, and worst-case space complexities**

Finally, represent the comparative results **graphically by plotting the complexity values on a two-dimensional (2D) plane**, and critically analyze the plots to justify the suitability of each method for large-scale census data processing.

3. A **financial technology analytics firm** processes transaction data from two independent banking systems. Each bank stores the **daily transaction amounts** of its customers in a **sorted array** (sorted in non-decreasing order). At the end of each day, the firm must compute the **median transaction value** across both banks to generate risk and liquidity assessment reports. Due to strict performance requirements and memory constraints, the firm cannot afford to merge the two arrays into a single array. Instead, the median must be computed directly from the two sorted datasets using an **efficient algorithm**.

Write a C program to compute the median of the combined dataset formed by two given sorted arrays, without merging the arrays into a single structure. Further, examine whether it is possible to design an algorithm that solves this problem with **logarithmic time complexity**. If such an algorithm exists, implement your solution strictly following that approach and justify its efficiency through appropriate time complexity analysis.